

CMAS: A Cross-Modal Attribution Score for Modality-Level Explanation Evaluation in Audio-Visual Deepfake Detection

Tejvir Singh Grewal

Department of Computer Science and Engineering, Thapar Institute of Engineering and Technology

tgrewal_be23@thapar.edu

Abstract

Multimodal deepfake detectors combine visual and audio information, but their evaluation usually focuses on whether the final REAL or FAKE prediction is correct. That leaves a separate question: when a detector explains a fake clip, does the explanation point to the modality that was actually manipulated? FakeAVCeleb provides a useful setting for this question because its clips include labels for video-only, audio-only, and both-modality manipulations. We introduce the Cross-Modal Attribution Score (CMAS), a simple metric that compares a two-element visual and audio attribution vector with the corresponding manipulated-modality label using cosine similarity. We evaluate CMAS with a multimodal detector built from an EfficientNet-B0 visual encoder, a Wav2Vec2-base audio encoder, and a cross-attention fusion module. On the final 3,103-clip test split, the multimodal model reaches 99.55% accuracy and 99.77% F1. The visual-only and audio-only baselines reach 97.23% and 84.56% accuracy, with F1 scores of 98.56% and 91.49%, respectively. For the reported CMAS analysis, modality-ablation attribution has a mean score of 0.9997 and Integrated Gradients has a mean score of 0.9466 on 125 FAKE samples with defined manipulated-modality labels. We also document a calibration problem found during development: balanced sampling was used together with inverse-frequency loss weighting, which placed the decision threshold poorly for the initial models. Removing the extra loss weighting and retraining the audio-only and multimodal models produced better-calibrated default predictions. The results are presented with the limitations of the dataset, the final split, and the CMAS sample size stated explicitly.

Keywords: deepfake detection, multimodal learning, audio-visual fusion, explainable AI, attribution methods, cross-modal explanation faithfulness, FakeAVCeleb

1. Introduction

Synthetic media has become easier to create and harder to assess by visual inspection alone. Face swapping, lip-sync manipulation, and voice cloning can produce clips that look or sound plausible to a viewer. This has increased interest in automated methods that can distinguish real and manipulated content.

A common way to evaluate a deepfake detector is to report accuracy, F1, or ROC-AUC on a held-out test set. These metrics are important, but they describe the final decision rather than the evidence used to reach it. A model can perform well on a benchmark while relying on cues that are correlated with the dataset or its generation process. Explanation methods can help inspect model behavior, but a plausible explanation is not automatically a faithful one. In a multimodal setting there is an additional question: if only the audio was manipulated, does the model's explanation place more importance on audio, and does the same hold for video-only and both-modality manipulations?

Methods such as Grad-CAM, SHAP, and Integrated Gradients are often used to show which parts of an input influence a prediction. Some studies also evaluate explanations by perturbing the highlighted regions and measuring how the prediction changes. These approaches are useful for studying faithfulness to the model's decision, but they do not directly compare an explanation with the known source of a manipulation. A modality-level comparison is possible when a dataset records which stream was altered.

FakeAVCeleb is useful for this purpose because it provides both a REAL or FAKE label and a manipulated-modality label for each clip. The latter identifies video-only, audio-only, and both-modality manipulations. This additional label makes it possible to evaluate whether a multimodal explanation agrees with the known manipulated stream.

This paper introduces the Cross-Modal Attribution Score (CMAS). CMAS compares a model's visual and audio attribution values with the manipulated-modality label in FakeAVCeleb. We use a multimodal detector with EfficientNet-B0 and Wav2Vec2-base encoders and compare it with visual-only and audio-only versions of the same system. Two attribution methods are evaluated: leave-one-modality-out ablation and Integrated Gradients at the embedding level. The goal is not to claim a new state-of-the-art detector, but to provide a direct and reproducible way to assess modality-level attribution on a dataset that provides the required ground truth.

The contribution is deliberately narrow. CMAS is intended as a modality-level evaluation measure, not as a complete test of explanation faithfulness. A high CMAS score shows agreement between the attribution vector and the dataset's manipulated-modality label. It does not prove that the model found the exact manipulation artifact, and it does not establish causal dependence on that artifact. The experiments are also limited to one dataset and one detector architecture. These limits are important when interpreting the results.

2. Related Work

2.1 Audio-visual deepfake detection

Audio-visual deepfake detection has developed from separate visual and audio detectors toward models that combine information from both streams. FakeAVCeleb was designed for this setting and includes labels identifying which modality was manipulated. Recent work has explored cross-modal fusion and attention-based approaches. Oorloff et al. [7] introduced AVFF, which uses audio-visual feature fusion and reports strong results on FakeAVCeleb. Katamneni and Rattani [8] proposed a contextual cross-modal attention approach for detection and localization. Koutlis and Papadopoulos [10] explored modality information differences at the discourse level, while Liu et al. [9] reviewed the development from single-modal to multimodal facial deepfake detection. More recent benchmarking work has also raised concerns about shortcuts in FakeAVCeleb and the reliability of standard evaluation protocols [11]. These studies motivate the use of both streams, while also showing why benchmark results need to be interpreted carefully.

2.2 Explainability for deepfake detection

Grad-CAM, SHAP, LIME, and Integrated Gradients [4] have been used to make deepfake detectors more interpretable. Tsigos et al. [5] studied quantitative evaluation of explanation methods for deepfake detection using perturbation-based measures of how explanations affect model confidence. Bharati et al. [6] presented an explainable deepfake detection framework aimed at producing human-interpretable rationales for legal investigation. These studies show the value of evaluating explanations rather than presenting saliency maps alone. CMAS asks a different and narrower question: when the dataset provides a manipulated-modality label, does the model's modality-level attribution agree with that label?

2.3 Positioning

The distinction from earlier work is therefore the evaluation target. CMAS compares a two-element attribution vector directly with the ground-truth manipulated-modality label supplied by FakeAVCeleb. We are not claiming that cosine similarity is a complete measure of explanation quality, nor that CMAS replaces perturbation-based or human-centered evaluation. The claim made here is limited to this specific formulation and its evaluation on the present dataset and model.

3. Problem Formulation

We consider binary classification of short audio-visual clips into REAL or FAKE. Each clip additionally carries, at training and evaluation time only (never as a model input), a manipulated-modality label $m \in$

{none, video, audio, both}, indicating which part of the original real clip was altered to produce the sample being labeled. This label exists in FakeAVCeleb by construction, since the dataset's generation pipeline records exactly which manipulation method (if any) was applied to each stream.

The detector itself is a standard function f that maps a clip's visual and audio streams to a REAL/FAKE prediction; nothing about f 's training objective uses the modality label. Separately, an explanation method g maps the same clip to a two-element vector $e = [e_{\text{visual}}, e_{\text{audio}}]$, intended to represent the relative contribution of each modality to the model's prediction. The question addressed here is not only whether f is accurate, but also whether e agrees with the true manipulated modality for clips with a defined modality label. Section 4.2 makes this comparison precise.

4. Proposed Method

The proposed method has three parts: the multimodal detector architecture (4.1), the CMAS formulation used to evaluate its explanations (4.2), and the two attribution methods CMAS is computed against (4.3).

4.1 Multimodal Architecture

The detector has three components: a visual encoder, an audio encoder, and a fusion-plus-classification head. All three are trained jointly for the multimodal configuration; the visual-only and audio-only ablations reuse the same encoder and simply route its output directly to a classification head, bypassing fusion.

4.1.1 Visual encoder

Frames are sampled from each clip via OpenCV at fixed intervals (8 frames per clip in our configuration), and a face region is cropped from each sampled frame using MediaPipe's face-detection model. The cropped faces are passed through an EfficientNet-B0 [2] backbone, initialized from ImageNet-pretrained weights, and the resulting per-frame feature vectors are averaged into a single 1280-dimensional visual embedding per clip. EfficientNet-B0 was chosen for its favorable accuracy-to-compute ratio, which matters in a setting (a single-author capstone project run on Colab-class GPUs) where training-time budget is a genuine constraint rather than an afterthought.

4.1.2 Audio encoder

Audio is extracted from each clip via ffmpeg, resampled to 16kHz mono, and passed through Wav2Vec2-base [3], a self-supervised speech representation model pretrained on unlabeled speech audio. We take the mean of the model's final hidden-state sequence as a single 768-dimensional audio embedding per clip, rather than fine-tuning a task-specific head on top of the full sequence, again for compute-budget reasons.

4.1.3 Cross-attention fusion

The multimodal configuration combines the two embeddings via a bidirectional cross-attention layer: the visual embedding attends over the audio embedding and vice versa, the two attended representations are concatenated, and a small feed-forward classification head produces the final REAL/FAKE logits. One architectural detail is worth stating plainly because it directly affects how the explainability results in Section 7.2 should be read: because each modality is reduced to a single pooled embedding before fusion, the cross-attention operation attends over exactly one key per modality. A softmax over a single key is mathematically forced to output 1.0 regardless of the query, which we verified directly rather than assumed.

This means the raw attention weights inside the fusion layer carry no usable signal about modality importance. We discovered this during development, when an early attempt to use those weights as an explanation method produced a constant, input-independent result. Section 4.3 describes the two attribution methods we use instead, neither of which depends on the raw attention weights.

4.2 CMAS Formulation

For a clip with manipulated-modality label m , we define a ground-truth modality vector g as follows:

$$g = [1, 0] \text{ if } m = \text{video}; \quad g = [0, 1] \text{ if } m = \text{audio}; \quad g = [0.5, 0.5] \text{ if } m = \text{both}$$

Clips with $m = \text{none}$ (unmanipulated REAL clips) have no manipulated modality by definition, so g is undefined for them, and such clips are excluded from CMAS computation entirely. CMAS is only meaningful for a clip where a true modality-level answer exists to compare against.

Given an explanation method that produces a raw, non-negative two-element importance vector for a clip, we normalize it to the unit simplex ($e = [e_{\text{visual}}, e_{\text{audio}}]$, $e_{\text{visual}} + e_{\text{audio}} = 1$, both non-negative), falling back to the uninformative $[0.5, 0.5]$ if the raw vector is degenerate (all zero). CMAS for a single clip is then the cosine similarity between e and g :

$$\text{CMAS}(\text{clip}) = (e \cdot g) / (\|e\| \|g\|)$$

Since e and g contain only non-negative values, their cosine similarity lies between 0 and 1. For video-only and audio-only samples, a score of 1 means that the attribution points fully to the manipulated modality, while a score of 0 means that it points fully to the other modality. The interpretation is different for both-modality samples because $g = [0.5, 0.5]$. In that case, an attribution of $[0.5, 0.5]$ gives a score of 1, while an attribution concentrated entirely on one modality gives a score of $1/\sqrt{2}$, approximately 0.707. Therefore, a lower CMAS value does not have the same meaning across all manipulation types. We report the mean and standard deviation over the evaluated clips, restricted to samples predicted as FAKE and having a defined manipulated-modality label.

CMAS is deliberately a coarse, clip-level, modality-level metric. It does not attempt to localize which frames or which audio segment were responsible for a manipulation, only which stream. We return to this scope limitation directly in Section 9.

4.3 Explainability Methodology

We compute the raw importance vector two ways, and report both, since they make different tradeoffs between cost and faithfulness.

4.3.1 Modality-ablation importance

The cheaper method measures how much the model's predicted-class logit changes when each modality's embedding is independently zeroed out before fusion: `importance_visual` is the drop in the target logit caused by zeroing the visual embedding, and `importance_audio` the corresponding drop from zeroing the audio embedding. Both are clamped to non-negative (a modality whose removal increases the target logit contributed nothing positive to it) and normalized to sum to 1. This method is architecture-agnostic: it depends only on being able to re-run the small fusion-and-classification head with one input zeroed, not on any particular fusion mechanism. Unlike the raw cross-attention weights discussed in Section 4.1.3, we verified empirically that it produces genuinely input-dependent output rather than a constant.

4.3.2 Integrated Gradients

The second method is Integrated Gradients [4], applied at the embedding level: we attribute the predicted-class logit back to the visual and audio embedding vectors (rather than to raw pixels or waveform samples, which would require a different and more expensive integration path), using a zero baseline and 50 interpolation steps, and reduce each modality's attribution to a scalar via the sum of absolute per-dimension attribution. This is more expensive per clip than the ablation method, since it requires multiple forward-and-backward passes rather than two extra forward passes, but it is a more standard, axiomatically-motivated attribution technique, and we report it specifically to see whether the two methods agree.

A third row we report, labeled "both," is the simple average of the two normalized importance vectors before computing CMAS. This is not a claim that averaging is the theoretically correct way to combine the two signals; it is included because it is what a practitioner reaching for "the model's explanation" without picking a specific method by name would likely compute, and it is useful to know how CMAS behaves under that default choice.

5. Dataset

We use the full FakeAVCeleb release [1], containing 21,544 clips. The dataset contains real and manipulated clips created using face-swap and voice-cloning procedures, and each clip has a REAL or FAKE label together with a manipulated-modality label. We independently checked the downloaded release and found 500 real clips, 9,709 video-only manipulated clips, 500 audio-only manipulated clips, and 10,835 both-modality manipulated clips. This corresponds to a global FAKE fraction of approximately 97.68%. The relatively small number of real clips is important when interpreting metrics for the REAL class.

Train, validation, and test manifests were created using an identity-disjoint split of approximately 70/15/15. Clips were grouped by the source-identity folder so that the same underlying identity was not intentionally placed in both training and evaluation sets. The final evaluation reported in this paper uses a 3,103-clip test split. During development, rerunning the identity-based manifest generation produced test splits of approximately 3,155 to 3,103 clips. The final split was used consistently for the reported model results. We also scanned the referenced video files and found no corrupted files among the 21,544 clips.

The dataset imbalance affects the interpretation of the results. Real clips and audio-only manipulated clips each account for only 500 samples in the full release, while video-only and both-modality clips make up most of the dataset. In the final test split, only 75 clips are REAL. As a result, REAL-class metrics are based on a small number of examples and can change noticeably when a single clip is misclassified.

6. Experimental Setup

All three models (visual-only, audio-only, multimodal) were trained with AdamW, mixed-precision training, a cosine learning-rate schedule with linear warmup, and discriminative learning rates (a lower rate for the pretrained backbones, a higher rate for the newly initialized fusion and classification layers). Backbones were frozen for the first three epochs and unfrozen afterward, so that the randomly initialized classification head could stabilize before backpropagating through the much larger pretrained encoders. Early stopping monitored validation F1 with a patience of five epochs. Training and evaluation were run on Google Colab GPU instances.

6.1 A calibration failure we found and fixed

The first training configuration combined balanced batch sampling with inverse-frequency class weighting. Because the dataset is heavily imbalanced toward FAKE clips, these two mechanisms both compensated for the same class imbalance. In the initial runs, the models produced high precision for the FAKE class but much lower FAKE recall at the default decision threshold, while ROC-AUC remained substantially higher. This indicated that the score ranking was more useful than the default operating point suggested.

We checked this behavior in several ways. The class balance implied by the initial predictions was close to the measured dataset balance. A validation-based Youden's-J threshold also improved the visual-only model's F1 from the original default-threshold result without changing its weights. We then removed inverse-frequency loss weighting while retaining balanced sampling and retrained the audio-only and multimodal models. Those two retrained models no longer showed the same default-threshold behavior. The visual-only model was not retrained, so it is reported separately with both its default-threshold and validation-tuned results.

This experience is included because it affected the evaluation and is useful for interpreting the final numbers. Balanced sampling and class-weighted loss can each be reasonable choices for an imbalanced problem, but using both requires checking how they affect the model's output calibration. The final audio-only and multimodal results use the retrained models after the extra class weighting was removed.

6.2 Baselines

We compare three configurations of the same underlying architecture rather than comparing against external published baselines, since the point of the comparison is to isolate the contribution of fusion, not to claim state-of-the-art detection accuracy on FakeAVCeleb:

- Visual-only: EfficientNet-B0 encoder and classification head, no audio input.
- Audio-only: Wav2Vec2-base encoder and classification head, no visual input.
- Multimodal: both encoders plus cross-attention fusion, as described in Section 4.1.

This is a within-architecture ablation, not a benchmark comparison against prior published detectors; Section 9 discusses this scope limitation directly.

6.3 Evaluation Metrics

We report accuracy, precision, recall, F1, and ROC-AUC, with FAKE treated as the positive class. The default operating point is the 0.5 threshold, or the class selected by the larger output logit. For completeness, validation-tuned Youden's-J thresholds and the resulting test metrics are also available from the evaluation runs. The main comparison in Table 1 uses the reported threshold for each model as described in the table footnote, so the visual-only exception is explicit rather than hidden.

7. Results

Table 1 reports detection performance for all three models on the 3,103-clip test split. The operating point for each model is described in the table footnote.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Visual-only	97.23% ¹	99.93% ¹	97.23% ¹	98.56% ¹	98.36%
Audio-only	84.56%	99.04%	85.01%	91.49%	83.60%
Multimodal	99.55%	99.97%	99.57%	99.77%	99.93%

Table 1. Detection performance on the FakeAVCeleb test split ($n = 3,103$).

¹ The visual-only model was trained before the calibration fix described in Section 6.1 and was not retrained; its Accuracy, Precision, Recall, and F1 figures use its validation-tuned decision threshold (0.0066), since its default-threshold performance still shows the original miscalibration symptom (61.6% accuracy, 76.3% F1, at identical ROC-AUC). The audio-only and multimodal figures use the untouched default threshold (0.5), since both were retrained after the fix and did not benefit from threshold tuning. ROC-AUC is threshold-independent throughout.

The multimodal model gives the strongest results among the three configurations in the final evaluation. It reaches 99.55% accuracy and 99.77% F1, compared with 97.23% accuracy and 98.56% F1 for the visual-only model and 84.56% accuracy and 91.49% F1 for the audio-only model. The comparison is useful as an internal ablation because all three systems use the same general encoder setup and differ in whether one or both modalities are available.

At the default threshold, the multimodal confusion matrix is $\begin{bmatrix} 74 & 1 \\ 13 & 3015 \end{bmatrix}$, with rows corresponding to true REAL and true FAKE and columns corresponding to predicted REAL and predicted FAKE. This gives one false positive and thirteen false negatives in the 3,103-clip test split. Only 75 test clips are REAL, so the REAL-class result should be interpreted with caution despite the high overall accuracy.

7.1 Ablation Study

Table 1 also serves as the main ablation study. The visual-only model uses EfficientNet-B0, the audio-only model uses Wav2Vec2-base, and the multimodal model uses both encoders with the fusion module. The three configurations share the same general training and evaluation pipeline, with the visual-only model being the exception in that it was not retrained after the calibration change. The ranking is consistent across the reported metrics, with the multimodal configuration performing best and the audio-only configuration performing worst on this test split.

Figures 1 and 2 show the ROC curve and confusion matrix for the audio-only model. They provide a visual view of its weaker performance and show that its ROC-AUC of 0.836 is considerably better than its default-threshold accuracy might suggest.

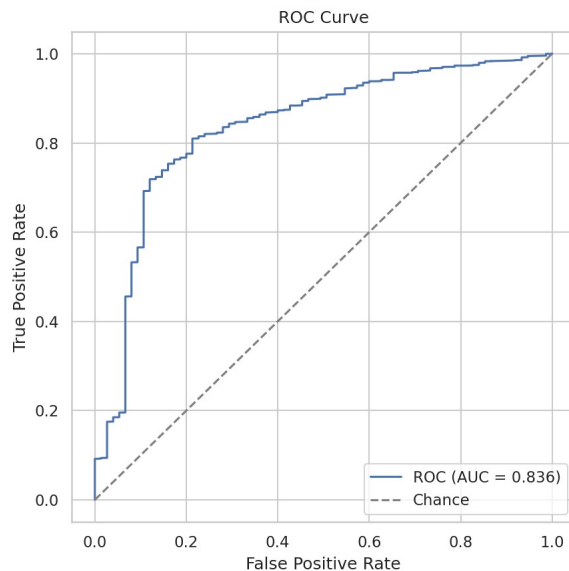


Figure 1. ROC curve, audio-only baseline (test split, AUC = 0.836).

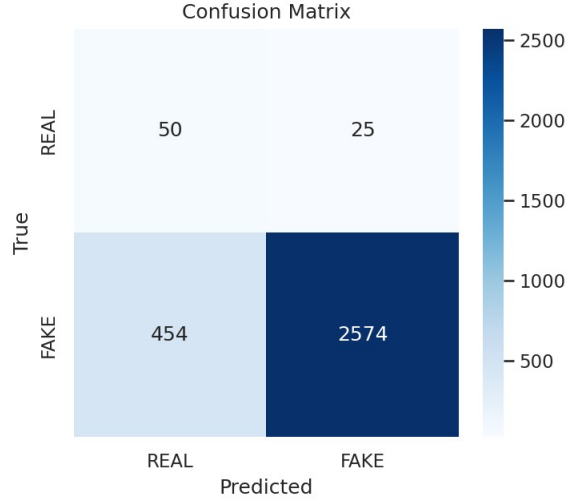


Figure 2. Confusion matrix, audio-only baseline, default threshold (test split).

7.2 Explainability Analysis

Table 2 reports the CMAS results for the multimodal model on 125 FAKE samples with a defined manipulated-modality label. The CMAS experiment processes the first 200 samples from the test loader. This limit was used because attribution computation, particularly Integrated Gradients with 50 interpolation steps, is substantially more expensive than standard inference. The test loader uses a fixed order with shuffling disabled, so the evaluated subset is the first 200 samples in the saved test manifest. Of these 200 samples, 75 are REAL and are excluded because no manipulated modality is defined for them, leaving 125 FAKE samples for CMAS aggregation. The reported CMAS values therefore describe this evaluated subset rather than the full test split.

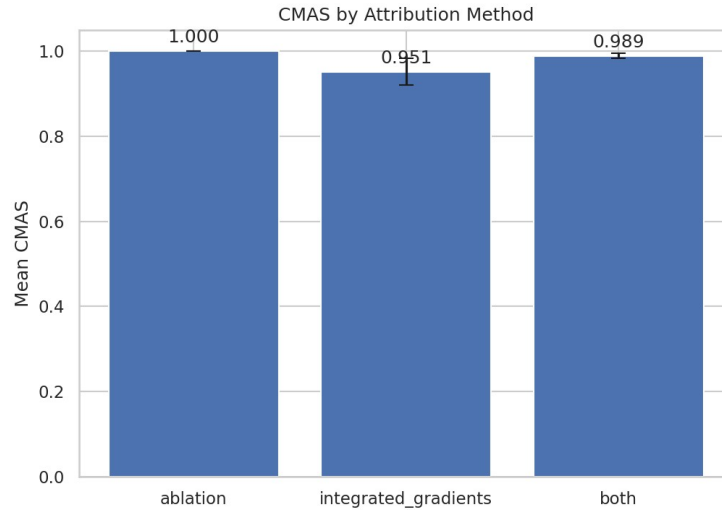


Figure 3. CMAS by attribution method, multimodal model (test split).

Attribution method	Mean CMAS	Std. dev.
Ablation (leave-one-modality-out)	0.9997	0.0052
Integrated Gradients	0.9466	0.0340
Both (averaged)	0.9859	0.0067

Table 2. CMAS by attribution method, multimodal model (125 FAKE samples from the first 200 test samples).

The reported CMAS values are high for both attribution methods, with a higher mean for modality ablation than for Integrated Gradients. Within the evaluated subset, the modality-level attributions generally agree with the manipulated-modality labels in FakeAVCeleb. Because the analysis is limited to the first 200 test samples, the result should not be treated as a dataset-wide estimate of explanation agreement.

The difference between the two methods is also useful to note. Modality ablation asks how much the predicted-class logit changes when one modality is removed, while Integrated Gradients assigns attribution along a path from a zero embedding baseline to the observed embedding. In this experiment, the ablation scores have a higher mean and lower standard deviation. This may reflect the fact that the ablation test asks a simpler modality-level question. It is not enough evidence to conclude that one attribution method is generally better than the other.

Figure 4 shows the visual and audio contribution values for six individual FAKE samples using the averaged attribution vector. The examples illustrate that the attribution is not fixed to one constant visual-to-audio ratio across the displayed clips. They are included as qualitative examples and are not intended as a statistical summary of the test set.

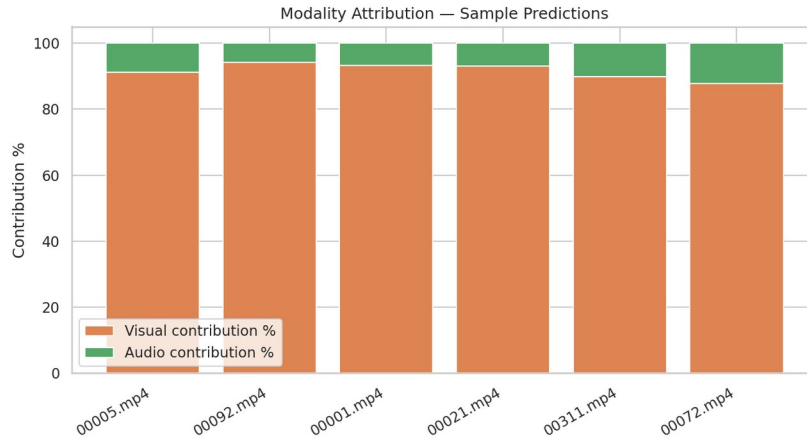


Figure 4. Visual versus audio contribution, six individual test-split FAKE samples (multimodal model).

7.3 Error Analysis

The multimodal model makes thirteen false-negative errors on FAKE clips and one false-positive error on REAL clips in the final test split. This error count is too small for a reliable pattern analysis. The audio-only model has 454 false negatives at the default threshold, so its larger error set gives a clearer indication of its main weakness on this dataset: it misses many manipulated clips.

8. Discussion

The results support two practical observations. First, the initial training configuration showed that good ROC-AUC does not guarantee a useful default decision threshold when class imbalance is handled in more than one way. The visual-only model's F1 improved substantially when its threshold was tuned on validation data, while removing the extra class weighting and retraining the audio-only and multimodal models produced better-calibrated default predictions. Second, the CMAS analysis shows strong agreement between the model's modality-level attributions and FakeAVCeleb's manipulated-modality labels within the evaluated subset.

CMAS should not be interpreted as proof that the model is causally responding to the manipulation itself. A high score only means that the attribution vector agrees with the dataset label at the modality level. The model could still rely on a correlated cue inside the correct modality. Establishing that distinction would require more detailed spatial or temporal attribution experiments. The present results therefore support a limited claim about modality-level agreement, not a complete claim about causal or artifact-level faithfulness.

9. Limitations

Several limitations bound how far these results should be read.

- Single dataset and single architecture. All experiments use FakeAVCeleb and one encoder-and-fusion design. The results do not establish generalization to other datasets, manipulation methods, or architectures.
- No external detector comparison. The visual-only, audio-only, and multimodal models form an internal ablation. We do not claim state-of-the-art performance against published detectors because those systems were not reimplemented and evaluated under the same conditions.
- Small REAL-class sample. The full dataset contains 500 real clips, and only 75 are present in the final test split. Metrics for the REAL class therefore have greater uncertainty than the aggregate FAKE-class results.
- Test-split drift during development. Rerunning the identity-based manifest generation produced test sets of approximately 3,155 to 3,103 clips. All final results in this paper use the 3,103-clip split. The split should be saved and treated as fixed for any future reproduction of these numbers.

Recent benchmarking work has identified a leading-silence shortcut in FakeAVCeleb [11]. We did not run a shortcut-controlled evaluation in this study, so the near-perfect detection scores reported here should not be interpreted as evidence of robust generalization beyond the benchmark conditions.

- CMAS is a modality-level, not a spatial or temporal, metric. It answers "which modality" and cannot currently answer "which frame" or "which second of audio." Section 10 discusses what a finer-grained extension would require.
- Degenerate raw attention weights. Because each modality is reduced to one pooled embedding before cross-attention, each attention operation has only one key. Its softmax weight is therefore 1.0 by construction, so the raw attention weights cannot be used as a meaningful modality attribution. We instead use modality ablation and Integrated Gradients.
- Visual-only baseline was not retrained after the calibration change. Its reported accuracy, precision, recall, and F1 use a validation-tuned threshold, while the audio-only and multimodal results use their default threshold after retraining. This makes the comparison useful as an internal reference, but not perfectly controlled.

10. Future Work

The most direct extension is a multi-token cross-attention fusion design (per-frame visual tokens attending over per-chunk audio tokens, rather than each modality collapsing to one pooled vector before fusion), which would make the fusion layer's own attention weights a legitimate, non-degenerate attribution signal, and would likely allow CMAS-style evaluation at finer than modality granularity. A second direction is evaluating CMAS across additional audio-visual deepfake datasets and additional fusion architectures, to establish (or find the limits of) whether the strong agreement between explanation and ground truth we observed here holds more generally, or is specific to this architecture and this dataset's particular manipulation methods. A third, more modest direction is simply retraining the visual-only baseline under the corrected training configuration, so that all three ablation numbers in Table 1 are on equal footing rather than one relying on threshold tuning to reach comparable performance.

11. Conclusion

We introduced CMAS, a modality-level metric for comparing a multimodal detector's attribution with the manipulated-modality label available in FakeAVCeleb. We evaluated it with an EfficientNet-B0 and Wav2Vec2-base detector and compared multimodal, visual-only, and audio-only configurations. On the final 3,103-clip test split, the multimodal model reaches 99.55% accuracy and 99.77% F1, while the visual-only and audio-only models reach 97.23% and 84.56% accuracy, respectively. In the reported CMAS subset, modality ablation has a mean score of 0.9997 and Integrated Gradients has a mean score of 0.9466. These values indicate strong agreement between the evaluated modality attributions and the dataset labels, but the CMAS analysis is limited by its sample size and the use of a computationally manageable subset. We also recorded and corrected a training calibration issue involving the simultaneous use of balanced sampling and inverse-frequency loss weighting. The main value of the work is therefore not a claim of state-of-the-art detection, but a focused evaluation of whether a multimodal detector's explanations agree with known modality-level manipulation labels. Further evaluation on fixed splits, larger CMAS samples, other datasets, and other architectures would be needed to establish how well the metric generalizes.

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